



Loops

Simplifying Our Code

$$\frac{\overset{1}{\cancel{3}}}{\cancel{15}} \times \frac{\overset{1}{\cancel{5}}}{\cancel{6}} = \frac{1}{6}$$

The diagram shows the simplification of the fraction $\frac{3}{15} \times \frac{5}{6}$. The 3 in the numerator of the first fraction and the 15 in the denominator are crossed out with a red line. The 5 in the numerator of the second fraction and the 6 in the denominator are crossed out with a blue line. The result is $\frac{1}{6}$.

Purpose



Loops are extremely useful in programming to simplify our code.

They are an alternative to copy/paste programming.

Purpose



for loops

used to iterate over a set (i.e. lists, arrays, columns of data, etc.)

Example: printing all items in a list one-by-one

while loops

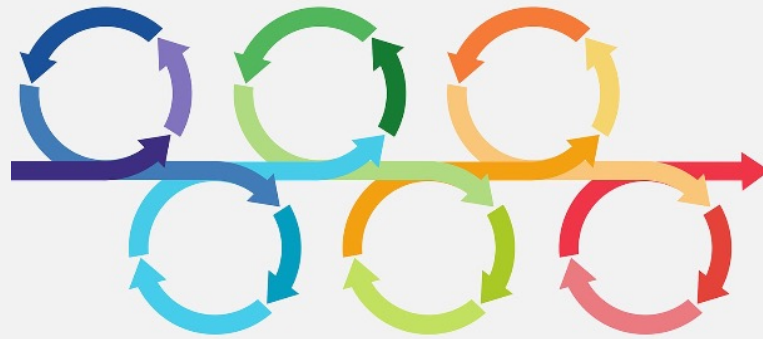
used to iterate over a variable

Example: giving a user three chances to guess a random number



For Loops

Iteration



Example: List Looping



If we create and print a list, we get the following output.

```
letter_lst = ['a', 'b', 'c', 'd', 'e']  
print( letter_lst )
```

Output:

```
['a', 'b', 'c', 'd', 'e']
```

Example: List Looping



To print each element of the list, we could do the following:

```
letter_lst = ['a', 'b', 'c', 'd', 'e']
```

```
print( letter_lst[0] )      # a
print( letter_lst[1] )      # b
print( letter_lst[2] )      # c
print( letter_lst[3] )      # d
print( letter_lst[4] )      # e
```



What if we had hundreds of elements?

What if the size of our list changes?

Example: List Looping



To print each element of the list, we could do the following:

```
letter_lst = ['a', 'b', 'c', 'd', 'e']
```

```
for letter in letter_lst:  
    print( letter )
```

Output:

```
a  
b  
c  
d  
e
```

For Loops



General syntax of a **for loop**

```
letter_lst = ['a', 'b', 'c', 'd', 'e']
```

```
for letter in letter_lst:  
    print( letter )
```

For Loops



The variable (letter) represents what we are **iterating over**.
It gets replaced with each **iteration**.

```
letter_lst = ['a', 'b', 'c', 'd', 'e']
```

```
for letter in letter_lst:  
    print( letter )
```

For Loops



The **iterable** we are **iterating** over.

```
letter_lst = ['a', 'b', 'c', 'd', 'e']
```

```
for letter in letter_lst:  
    print( letter )
```

For Loops with **break**



A **break** stops a loop if a condition is met.

```
letter_lst = [ 'a', 'b', 'c', 'd', 'e' ]
```

```
for letter in letter_lst:
```

```
    if letter == 'c':
```

```
        break
```

```
    print( letter )
```

Output:

a

b

For Loops with `continue`



A `continue` stops a loop if a condition is met.

```
letter_lst = ['a', 'b', 'c', 'd', 'e']  
  
for letter in letter_lst:  
    if letter == 'b':  
        continue  
    print( letter )
```

Output:

```
a  
c  
d  
e
```

Question



Which country has the best national football team?



Example: Argentina in the World Cup

Applying for loops





Python Practice

Applying for loops





While Loops

Conditional Iteration

Purpose



while loops

used to iterate over a variable

Example:

giving a user three chances to guess a random number

while Loops



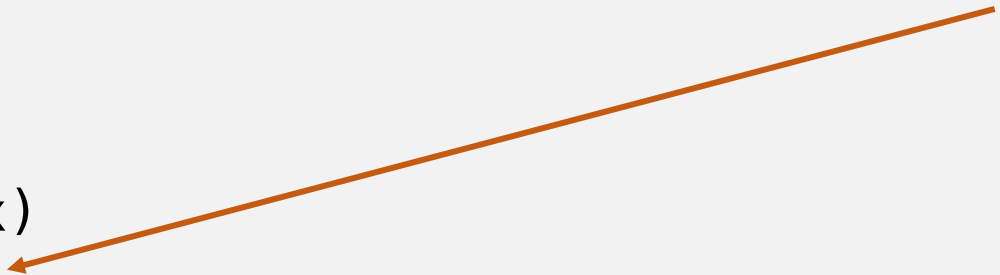
```
x = 5
```

```
while x > 1:
```

```
    print(x)
```

```
    x -= 1
```

This is very important!



Output:

5

4

3

2

Warning: DON'T DO THIS!!!!!!!!!!!!!!



```
x = 5
```

```
while x > 1:
```

```
    print(x)
```

Output:

```
5
5
5
5
5
5
5
```



Output:

5
5
5
5
5
5
5
5
5
5
5
5
5
5

Example: Space Shuttle Countdown



```
# Liftoff Countdown
```

```
count = 10
```

```
while count > 0:  
    print(count)  
    count -= 1
```

```
    input(prompt = "< Press ENTER to continue. >\n")
```

```
print("LIFTOFF!")
```

Example: Space Shuttle Countdown



Output

10

< Press ENTER to continue. >

9

< Press ENTER to continue. >

8

...

LIFTOFF!

Example: Guessing a Number Between 1 and 10



Build a while loop based on the following starter code.

Criteria

One conditional (if, elif, else statement)

```
import random
```

```
number = random.randint(a=1, b=10)
```

```
guesses = 3
```

```
import random
```

```
number = random.randint(a=1, b=10)
```

```
guesses = 3
```

```
while guesses > 0:
```

```
    choice = int(input(prompt = "> "))
```

```
    if choice == number:
```

```
        print("\nYou win!")
```

```
        break
```

```
    elif choice != number:
```

```
        print("\nThat's incorrect, please guess again.")
```

```
        guesses -= 1
```

```
    else:
```

```
        print("\nSomething went wrong.")
```

```
print(f"\nYou are out of guesses. The number was {number}.")
```

